

Dongyan Lin
dongyan.lin@mail.mcgill.ca
dongyanlin.github.io
+1 971-219-2979
LinkedIn: [dongyanlin](#)
GitHub: [dongyanlin](#)

Research Interests

I use insights from animal and human cognition to build more generalizable AI systems, and AI to accelerate scientific discovery in cognitive and neuroscience, across three threads: (1) **Human-inspired AI** – agents that learn like humans via *reinforcement learning*, *multimodal integration*, and *continual self-improvement*; (2) **AI for Science** – autonomous systems that overcome the cognitive biases LLMs inherit from training data to *generate novel hypotheses*; and (3) **Brain-AI Alignment** – shared principles between neural networks and brains, via *mechanistic interpretability* and representational geometry.

Education

Doctor of Philosophy – Computational Neuroscience

Mila & McGill University, Montreal, Canada

Sep 2019 – Sep 2024

Advisor: Blake Richards, GPA 4.0/4.0

Thesis: "Understanding the Capabilities and Constraints of Single-Neuron Tuning Curves with Deep Learning Models"

Honours Bachelors of Science – Physiology, Mathematics, Psychology

University of Toronto, Toronto, Canada

Sep 2015 – Jun 2019

Advisors: Steven Prescott, Katherine Duncan, Douglas Tweed, GPA 3.94/4.0

Research Positions

Postdoctoral Researcher – FAIR at Meta, New York, USA

Dec 2024 – Present

Working with [Emmanuel Dupoux](#) on the **Developmental AI** team, building **multimodal grounding** and **autonomous exploration** methods for language agents, including the [EgoBabyVLM](#) benchmark and studies comparing human and LLM causal reasoning.

Ph.D. Student – Mila & McGill University, Montreal, Canada

Sep 2019 – Sep 2024

Advised by [Blake Richards](#). Studied representation learning in brains and machines: trained **deep reinforcement learning** agents on simulated cognitive tasks, analyzed **calcium-imaging** data from mouse visual cortex and hippocampus, and built digital-twin CNN models of visual cortex to probe representational alignment.

Visiting Researcher – Berkeley AI Research (BAIR), California, USA

Jun 2024 – Sep 2024

Worked with [Alison Gopnik](#) on building **sample-efficient** RL agents that generalize to new environments by generating and testing **causal hypotheses**, inspired by causal learning in children.

Neuro-AI Intern – Cold Spring Harbor Laboratory, New York, USA

Jun 2022 – Sep 2022

Worked with [Tatiana Engel](#) on **transformer** architectures and hippocampal function, designing **reinforcement learning** agents that exploit latent structure – work that inspired the CCN 2023 GAC workshop reconciling neuron- and manifold-level views of neural computation.

Lab Manager / Research Assistant – University of Toronto, Toronto, Canada Sep 2018 – Apr 2019

Worked with [Katherine Duncan](#) in the Department of Psychology, running **VR** and **eye-tracking** experiments on semantic memory retrieval, conducting studies via Amazon MTurk, and managing lab operations.

Bioinformatics Intern – RIKEN Center for Integrative Medical Sciences, Yokohama, Japan Jun 2018 – Aug 2018

Worked with [Michiel de Hoon](#) using **genome arithmetic** and gene ontology methods to characterize non-coding RNA families and their evolutionary conservation in human THP-1 cells, contributing to a *Genome Research* publication.

Research Student – Sickkids Hospital, Toronto, Canada May 2017 – Aug 2017

Worked with [Steven Prescott](#) performing *in vitro* **extracellular recordings** on mouse hippocampal slices to study chloride regulation of gamma oscillations, and building physiological models in NEURON.

Publications

 [Google Scholar](#)

*: Equal contribution

◦: Core contributor

‡: mentee under my supervision

Papers

P12. D. Lin*, P. Rust*, A. Villar Corrales◦, A. W. M. Tan, M. Luthra, C. Saint-James, R. Moritz, S. Krogh-Jespersen, V. Stark, S. Parimi, J. Shen, Y. Bencheikroun, Y. Higuchi, M. Gleize, T. Fizycki, N. Hamilakis, M. Khentout, S. Tsuji, B. Kégl, J. Pino, M. C. Frank, E. Dupoux [EgoBabyVLM: Benchmarking Cross-Modal Learning from Naturalistic Egocentric Video Data](#) *ArXiv* (2026) [[Link to WIRED Article](#)]

P11. M. Samiei*, E. Yiu*, A. GX-Chen, **D. Lin**, J. Shen, B. A. Richards, A. Gopnik, D. Precup [Human Adults and LLMs as Scientists: Who Benefits from Active Exploration?](#) *Proceedings of the Annual Meeting of the Cognitive Science Society (CogSci)* (2026)

P10. M. Luthra*, J. Shen*, M. Poli*, A. Ortiz, Y. Higuchi, Y. Bencheikroun, M. Gleize, C. Saint-James, **D. Lin**, P. Rust, A. Villar, S. Parimi, V. Stark, R. Moritz, J. Pino, Y. LeCun, E. Dupoux [SpidR-Adapt: A Universal Speech Representation Model for Few-Shot Adaptation](#) *Annual Meeting of the Association for Computational Linguistics (ACL)* (2026)

P9. R. Algayres, C. Saint-James, M. Luthra, J. Shen, **D. Lin**, Y. Bencheikroun, R. Moritz, J. Pino, E. Dupoux. [LongTail-Swap: benchmarking language models' abilities on rare words](#). *Conference on Empirical Methods in Natural Language Processing (EMNLP)* (2025)

P8. A. GX-Chen, **D. Lin***, M. Samiei*, D. Precup, B. A. Richards, R. Fergus, K. Marino. [Language Agents Mirror Human Causal Reasoning Biases. How Can We Help Them Think Like Scientists?](#) *Conference on Language Modeling (CoLM)* (2025).

P7. A. GX-Chen*, **D. Lin***, M. Samiei*. [Testing causal hypotheses through Hierarchical Reinforcement Learning](#). *NeurIPS 2024 Workshop on Intrinsically-Motivated and Open-Ended Learning* (2024).

P6. D. Lin. [Fine-tuned network relies on generic representation to solve unseen cognitive task](#). *ICML 2024 Workshop on LLMs and Cognition* (2024).

P5. R. Tong, R. da Silva*, **D. Lin***, A. Ghosh*, J. Wilsenach, E. Cianfarano, P. Bashivan, B. A. Richards, S. Trenholm. [The feature landscape of visual cortex](#). *bioRxiv* (2023).

- P4. D. Lin**, A. Z. Huang[‡], B. A. Richards. [Temporal encoding in deep reinforcement learning agents](#). *Scientific Reports* (2023).
- P3. D. Lin**, D. Levenstein, J. H. Cornford, A. Ghosh, D. L. Barack, J. W. Krakauer, T. A. Engel, J. C. R. Whittington. [Neurons, circuits, or manifolds: reconciling the Sherringtonian and Hopfieldian views on neural computations](#). Generative Adversarial Collaborations (GAC) Proposal for 2023 Cognitive Computational Neuroscience (CCN) conference, Oxford, UK (2023).
- P2. M. de Hoon**, A. Bonetti, C. Plessy, Y. Ando, C-C. Hon, Y. Ishizu, M. Itoh, S. Kato, **D. Lin**, S. Maekawa, M. Murata, H. Nishiyori, J. W. Shin, J. Stolte, A. M. Suzuki, M. Tagami, H. Takahashi, S. Thongjuea, A. Forrest, Y. Hayashizaki, J. Kere, P. Carninci. [Deep sequencing of short capped RNAs reveals novel families of noncoding RNAs](#). *Genome Research* (2022).
- P1. D. Lin**, B. A. Richards. [Time cell encoding in deep reinforcement learning agents depends on mnemonic demands](#). *bioRxiv* (2021).

Conference Proceedings & Posters

- C8. D. Lin**, A. Z. Huang[‡], B. A. Richards. Time cells contribute to working memory through value-based recurrent dynamics. *Computational and Systems Neuroscience (COSYNE), Lisbon, Portugal. February 2024*. [\[Link to poster\]](#)
- C7. R. Tong**, A. Ghosh, E. Cianfarano, B. A. Richards, R. da Silva, **D. Lin**, J. Wilsenach, P. Bashivan, S. Trenholm. The feature landscape of visual cortex. *Computational and Systems Neuroscience (COSYNE), Lisbon, Portugal. February 2024*.
- C6. D. Lin**, R. da Silva, A. Ghosh, R. Tong, S. Trenholm, B. A. Richards. Neuronal optimal stimuli synthesized with deep learning reveal functional segregations in the mouse visual cortex. *Society for Neuroscience (SfN), San Diego, CA, USA. November 2022*. [\[Link to poster\]](#)
- C5. A. Z. Huang[‡], D. Lin**, B. A. Richards. Time cell encoding is decoupled from time perception in deep reinforcement learning agents. *Cognitive Computational Neuroscience (CCN), San Francisco, CA, USA. August 2022*. **Selected as a Contributed Talk (<5% of submissions)**. [\[Link to abstract\]](#) [\[Link to presentation\]](#)
- C4. D. Lin**, A. Z. Huang[‡], B. A. Richards. Heterogeneous Representations of Variables in Task-Optimized DRL Agents Depend on Task-Relevance. *The Multidisciplinary Conference on Reinforcement Learning and Decision Making (RLDM), Providence, RI, USA. June 2022*. [\[Link to paper\]](#)
- C3. D. Lin**, B. A. Richards. Time cell encoding in deep reinforcement learning agents depends on mnemonic demands. *Computational and Systems Neuroscience (COSYNE), Lisbon, Portugal. March 2022*. [\[Link to poster\]](#)
- C2. D. Lin**, FANTOM Consortium, M. de Hoon. Elucidating the functional roles of anti-sense transcripts in human THP-1 leukemia cells with computational methods. *Summer Undergraduate Research Symposium, Department of Cell & System Biology, University of Toronto. September 2018*. **Received the Best Poster Award**. [\[Link to poster\]](#)
- C1. D. Lin**, S. Ratté, S. A. Prescott. The chloride regulation of mouse hippocampal gamma oscillation in vitro. *University Research Opportunity Program Awardee Seminar, University of Toronto. July 2017*. [\[Link to presentation\]](#)

Awards & Honors

Michael Smith Foreign Study Supplement (for UC Berkeley)	\$6,000 (2024)
Society for Neuroscience Trainee Professional Development Award	\$1,000 (2022)
UNIQUE Centre Conference Travel Grant (for RLDM)	\$1,000 (2022)

FRQNT Doctoral Scholarship (declined to take CGS-D)	\$84,000 over 4 years (2022)
Alexander Graham Bell Canada Graduate Scholarship - Doctoral	\$105,000 over 3 years (2022)
Healthy Brains, Healthy Lives PhD Fellowship	\$15,000 over 1 year (2021)
IVADO MSc Excellence Scholarship	\$40,000 over 2 years (2020)
Integrated Program in Neuroscience Recruitment Award	\$5,000 (2019)
University of Toronto Dean's List	2019, 2018, 2017, 2016
Innis College Exceptional Achievement Award	\$735 (2018)
University of Toronto Research Opportunity Program Award	\$2,000 (2017)
Innis College Later Life Learning OSOTF Award	\$1,336 (2017)
Innis College Later Life Learning OSOTF Award	\$1,289 (2016)
University of Toronto Beatty Scholarship	\$1,500 (2016)
University of Toronto Entrance Scholarship	\$6,000 (2015)
Canada Governor General's Academic Bronze Medal	2015

Teaching

Teaching Assistant – Machine Learning

[IBRO-Simons Computational Neuroscience Imbizo](#), Cape Town, South Africa 2023, 2025

Teaching Assistant: INF8953DE (Reinforcement Learning)

Polytechnique Montréal, Montreal, Canada Fall 2021

Teaching Assistant / Project Manager

[AI4Good Lab](#), Montreal, Canada Summer 2021

Classroom Instructor

[BrainReach](#), Montreal, Canada Fall 2020, Winter 2021

Teaching Assistant: MAT135 (Calculus I), MAT136 (Calculus II)

University of Toronto, Toronto, Canada Fall 2018, Winter 2019

Classroom Instructor

Easy Group Inc., Toronto, Canada Sep 2016 – Apr 2018

Service

Program Committee

[Cognitive Computational Neuroscience \(CCN\)](#), New York City Oct 2025 – Aug 2026

Trainee Representative

[HBHL Equity, Diversity, and Inclusion Committee](#), Montreal, Canada Sep 2021 – Sep 2022

Mental Health First Aider

Mila, Montreal, Canada Feb 2021 – Sep 2024

Lab Representative

Mila, Montreal, Canada Nov 2021 – Nov 2022

Organizer

[Montréal AI & Neuroscience \(MAIN\) Conference](#), Montreal, Canada Oct 2020 – Nov 2020

Organizer

[UNIQUE Student Symposium](#), Montreal, Canada Feb 2020 – May 2020

Peer Mentor

[First-year Learning Community](#), University of Toronto, Toronto, Canada May 2018 – Apr 2019

Event Leader

[Centre for International Experience](#), University of Toronto, Toronto, Canada May 2018 – Apr 2019

Mentor

[Centre for International Experience](#), University of Toronto, Toronto, Canada May 2017 – Apr 2018

Paper Reviewing

NeurIPS 2021, 2022 [AI4Science Workshop](#), ICML 2022 AI4Science Workshop, ICLR 2023, ICML 2026, NeurIPS 2026

Skills

Programming Languages

Python, MATLAB, R, Bash/Shell, $\LaTeX$

Frameworks & Libraries

PyTorch, JAX, Hugging Face Transformers, NumPy, SciPy, pandas, scikit-learn, Gymnasium (OpenAI Gym), Suite2p, NEURON, OpenCV

Tools & Infrastructure

Git/GitHub, Unix/Linux, Slurm, Docker, W&B, Jupyter

Knowledge

Deep Reinforcement Learning, Representation Learning, Mechanistic Interpretability, Multi-modal & Vision-Language Models, Language Agents, Causal Reasoning, Computational & Systems Neuroscience, Calcium Imaging Analysis, Electrophysiology, Cognitive Modeling, Bioinformatics & Genomics

Natural Languages

English (fluent), Mandarin Chinese (native/fluent), French (intermediate)

Last updated: August 23, 2026